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Reference

video

book

OpenCourseWare - Mathematics for Computer Science - 6. Graph Theory & Coloring

Lecture Note

6. Graph Theory and Coloring

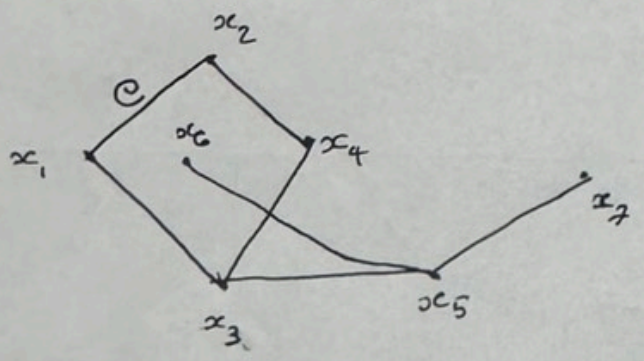
opposite gender problem

Claim (U Chicago): on average, men have 74% more opposite gender partners than women.

Claim (ABC News): " " 233% " " " "

nodes are connected edges (lines) which can be labeled

Def: A graph G is a pair of sets (V, E) where V is a nonempty set of items called vertices and nodes &
E is a set of 2-item subset of V called edges.



$V = \{x_1, x_2, \dots, x_7\}$

$E = \{\{x_1, x_2\}, \{x_1, x_3\}, \dots, \{x_5, x_7\}\}$

$x_1 - x_2$

order of edge doesn't mattered here

Ex:

$G = (V, E)$, $V = \{x_1, x_2, x_3\}$

x_1

x_2 x_3

x_i & x_j are adjacent if they are connected by an edge

if $\{x_i, x_j\} \in E$

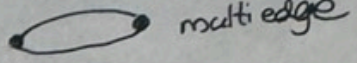
Def: Two nodes x_i & x_j

Def: An edge $e = \{x_i, x_j\}$ is incident to x_i & x_j

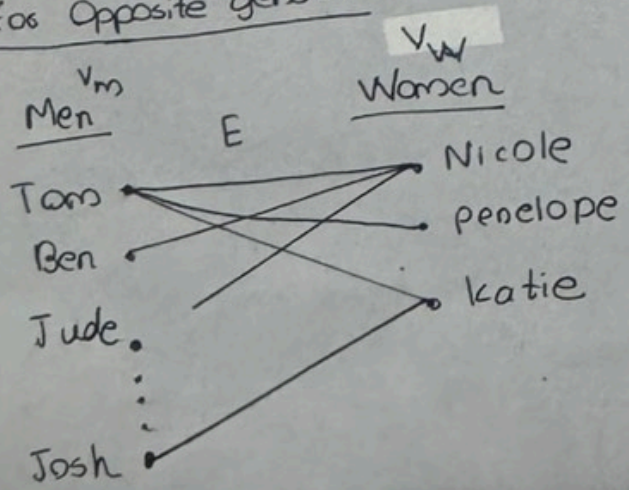
Def: The number of edges incident to a node is the degree of a node.

Def: A graph is a simple if has no loops or multiple edges





Graph for Opposite gender



$|V| \approx 300$ M

$|V_m| \approx 147.6$ M

$|V_W| \approx 152.4$ M

$|E| = ?$

Def: A_m = average # of opposite-geoder partner for men

A_W = " " " " " women

What is $\frac{A_m}{A_W} = ?$

$\sum_{x \in V_m} \deg(x) = E$

b.c it is just counting total edges which is E

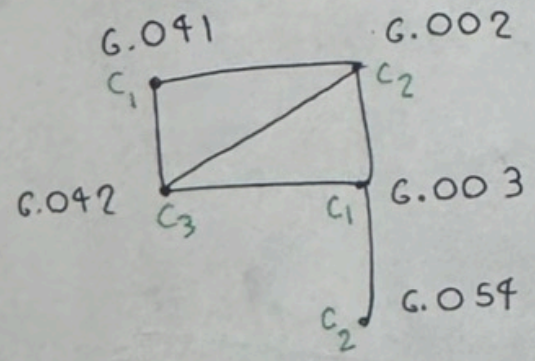
$$A_m = \frac{\sum_{x \in V_m} \deg(x)}{|V_m|} = \frac{|E|}{|V_m|}$$

$$A_W = \frac{\sum_{x \in V_W} \deg(x)}{|V_m|} = \frac{|E|}{|V_W|}$$

$$A_m / A_W = \frac{|E|/|V_m|}{|E|/|V_W|} = \frac{|V_m|}{|V_W|} = 1.0325$$ ← solution to opposite gender problem which don't need interview, survey

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Ex: exam schedule problem



goal: put slots into nodes

slots

C_1 Wed 5-7 pm

C_2 " 7-9 pm

C_3 " 9-11 pm

C_4 " 11-1 am

C_5 " 1-3 am

use as early time as possible

$\chi(G) = 3$

without nodes hooked by edge getting same slot

Graph Coloring problem: Given a graph G and k colors, assign a color to each node so that adjacent nodes get different color.

Def: The minimum value of k for which such a coloring exist is chromatic number of G

$\chi(G)$

Basic Coloring Alg for $G(V, E)$

1. order the nodes $v_1, \ldots, v_n$
2. order the color $c_1, c_2, \ldots, c_j$
3. for $i = 1, 2, \ldots, n$
   Assign the lowest legal color to v_i

→ greedy algorithm
always simple, work ok
work for all
give you upper bound

It become ordering problem

Thm: In every node in G has degree $\leq d$, then the basic algorithm use at most $d+1$ color for G.

Pf: By induction

$P(n)$: in every node in n node graph G. . . .

- for induction hypothesis in graph, you put (n) : number of nodes
- if doesn't work with n, put (e): number of edges

Base case: $n = 1 \Rightarrow$ 0 edges

- no degree (d)

$d = 0$ 1 color $= d+1$ ✓

Inductive step: Assume $P(n)$ is true for induction,
Let $G = (V, E)$ be any $(n+1)$ node graph
Let d = max degree in G

Order the nodes $v_1, v_2, \ldots, v_n, v_{n+1}$
Remove the $n+1$ from G to create $G' = (V', E')$

G' has the max degree $\leq d$ & n nodes so $P(n)$ say

Basic Alog use $\leq d+1$ colors for $v_1, v_2, \ldots, v_n$

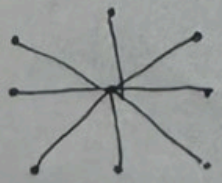
v_{n+1} has $\leq d$ neighbors $\Rightarrow$

$\exists$ color in $\{c_1, c_2, \ldots, c_{d+1}\}$ not used by any neighboars

Give v_{n+1} that color

$\Rightarrow$ Basic Alg uses $\leq d+1$ color on $G \Rightarrow P(n+1)$ ✓ □

Thinking worst case



$d = n-1$

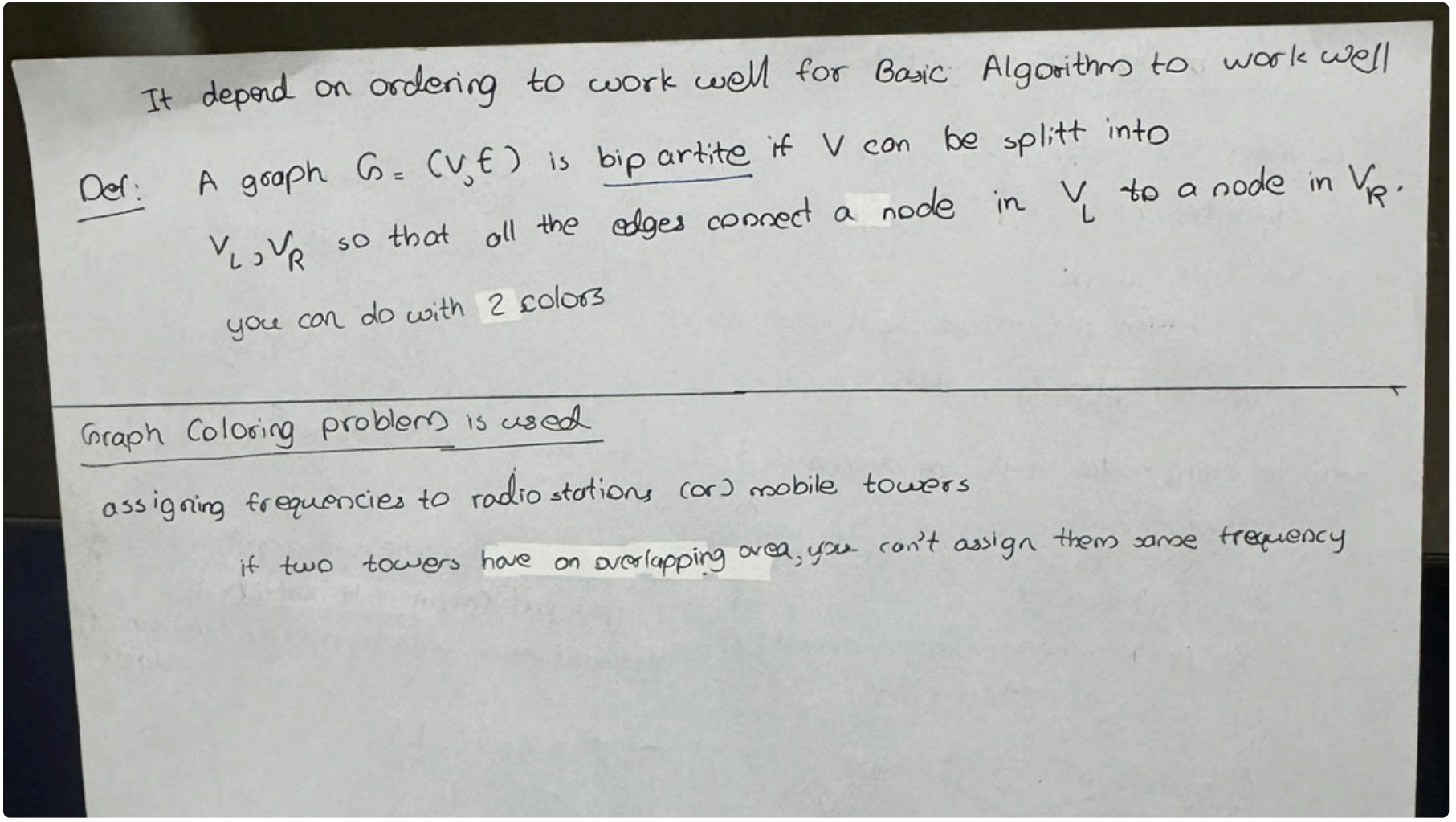
$\chi = 2$

upper bound from theorem $= n-1$

$\chi = 2$



complete bipartite



Book Reading : 5. Graph Theory

we will focus our attention on simple graphs where the relationship denoted by an edge is symmetric.

5.1 Definitions

5.1.1 Simple Graphs

Definition 5.1.1. A simple graph G consists of a nonempty set V , called the vertices (aka nodes) of G, and a set E of two-element subsets of V . The members of E are called the edges of G, and we write $G = (V, E)$.

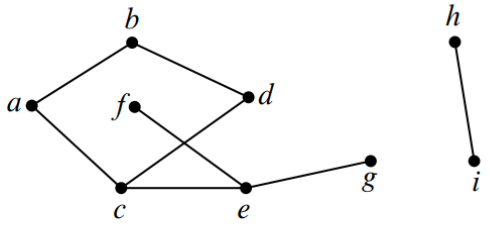


Figure 5.1 An example of a graph with 9 nodes and 8 edges.

$$V = \{a, b, c, d, e, f, g, h, i\}$$
$$E = \{\{a, b\}, \{a, c\}, \{b, d\}, \{c, d\}, \{c, e\}, \{e, f\}, \{e, g\}, \{h, i\}\}.$$

Note that $\{a, b\}$ and $\{b, a\}$ are different descriptions of the same edge, since sets are unordered. In this case, the graph $G = (V, E)$ has 9 nodes and 8 edges.

Definition 5.1.2. Two vertices in a simple graph are said to be *adjacent* if they are joined by an edge, and an edge is said to be *incident* to the vertices it joins. The number of edges incident to a vertex v is called the *degree* of the vertex and is denoted by $\deg(v)$; equivalently, the degree of a vertex is equals the number of vertices adjacent to it.

simple graphs doesn't contain

- self-loop
- multi-edges
- directed edges

5.1.2 Some Common Graphs

complete graph on n vertices, denoted K_n, has an edge between every two vertices, for a total of $\frac{n(n-1)}{2}$ edges.

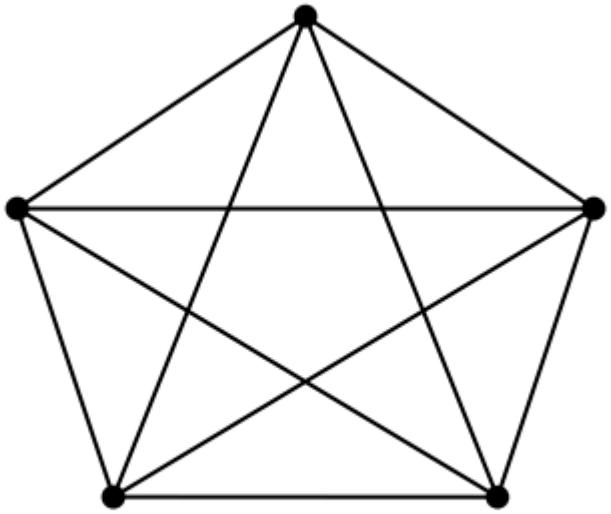


Figure 5.2 The complete graph on 5 nodes, K_5.

empty graph

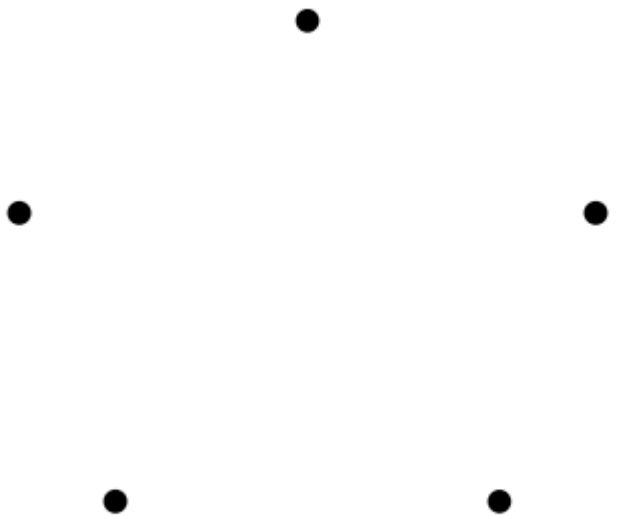


Figure 5.3 The empty graph with 5 nodes.

line graph L_n

The n-node graph containing $n-1$ edges in sequence is known as the *line graph* L_n. More formally, $L_n = (V, E)$ where

$$V = \{v_1, v_2, \ldots, v_n\}$$

and

$$E = \{\{v_1, v_2\}, \{v_2, v_3\}, \ldots, \{v_{n-1}, v_n\}\}$$

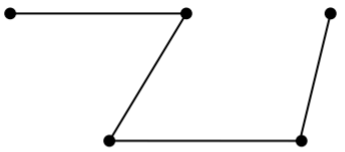


Figure 5.4 The 5-node line graph L_5.

cycle graph

If we add the edge $\{v_n, v_1\}$ to the line graph L_n, we get the graph C_n consisting of a simple cycle.

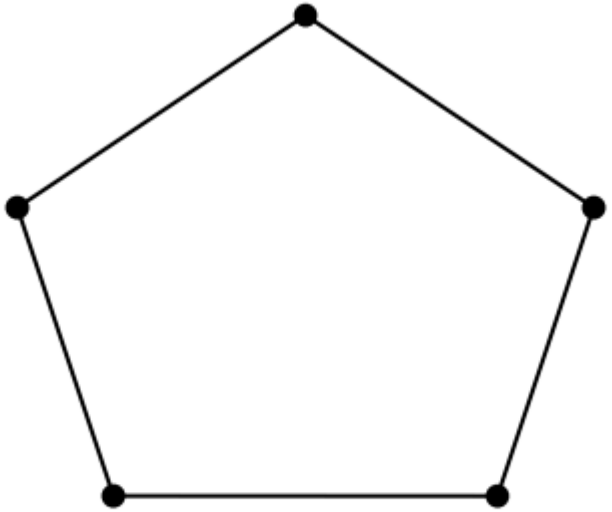


Figure 5.5 The 5-node cycle graph C_5.

5.1.3 Isomorphism

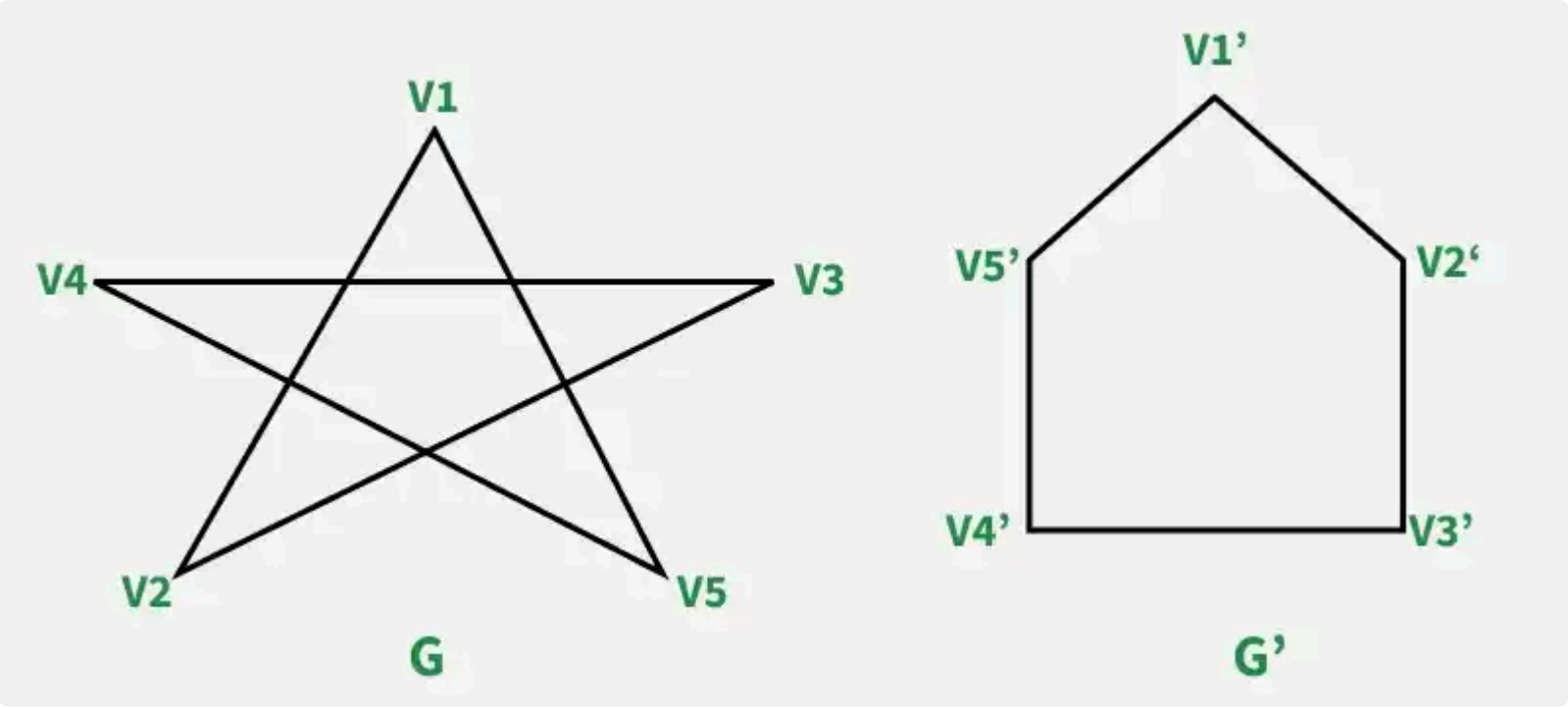
Definition 5.1.3. If $G_1 = (V_1, E_1)$ and $G_2 = (V_2, E_2)$ are two graphs, then we say that G_1 is *isomorphic* to G_2 iff there exists a *bijection*[4] $f : V_1 \to V_2$ such that for every pair of vertices $u, v \in V_1$:

$$\{u, v\} \in E_1 \quad \text{iff} \quad \{f(u), f(v)\} \in E_2.$$

The function f is called an *isomorphism* between G_1 and G_2.

Two graphs are isomorphic if:

- They have the same number of vertices
- They have the same number of edges
- The degree of corresponding vertices is the same
- The pattern of connections is identical (only labels may differ)



5.1.4 Subgraphs

5.1.4 Subgraphs

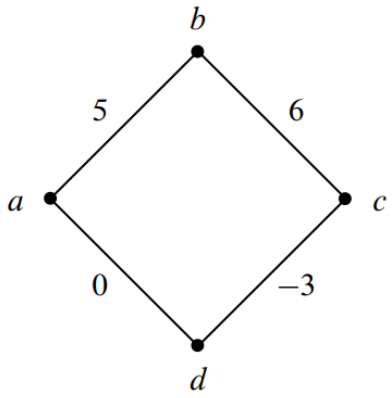
Definition 5.1.4. A graph $G_1 = (V_1, E_1)$ is said to be a *subgraph* of a graph $G_2 = (V_2, E_2)$ if $V_1 \subseteq V_2$ and $E_1 \subseteq E_2$.

For example, the empty graph on n nodes is a subgraph of L_n, L_n is a subgraph of C_n, and C_n is a subgraph of K_n. Also, the graph $G = (V, E)$ where

$$V = \{g, h, i\} \quad \text{and} \quad E = \{\{h, i\}\}$$

5.1.5 Weighted Graphs

A weighted graph is the same as a simple graph except that we associate a real number (that is, the weight) with each edge in the graph.



Mathematically speaking, a weighted graph consists of a graph $G = \{V, E\}$ and a weight function $w : E \to R$.

5.1.6 Adjacency Matrices

There are many ways to represent a graph.

- draw it
- sets
- adjacency metrics

Definition 5.1.5. Given an n-node graph $G = (V, E)$ where $V = \{v_1, v_2, \ldots, v_n\}$, the *adjacency matrix* for G is the $n \times n$ matrix $A_G = \{a_{ij}\}$ where

$$a_{ij} = \begin{cases} 1 & \text{if } \{v_i, v_j\} \in E \\ 0 & \text{otherwise.} \end{cases}$$

If G is a weighted graph with edge weights given by $w : E \to \mathbb{R}$, then the adjacency matrix for G is $A_G = \{a_{ij}\}$ where

$$a_{ij} = \begin{cases} w(\{v_i, v_j\}) & \text{if } \{v_i, v_j\} \in E \\ 0 & \text{otherwise.} \end{cases}$$

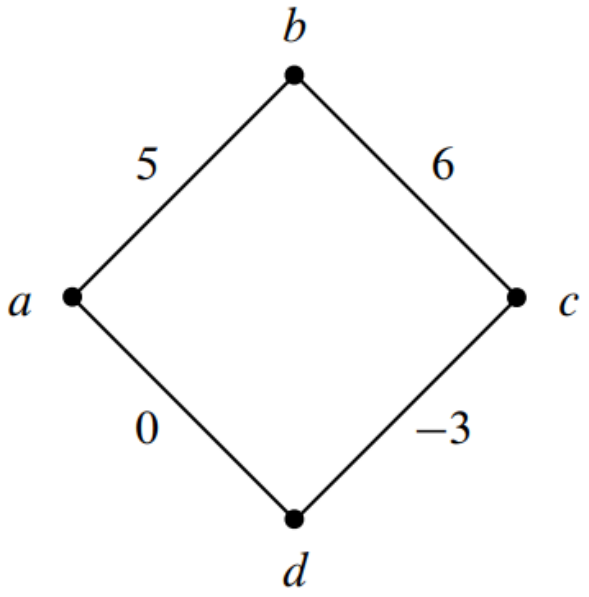


Figure 5.8 A 4-node weighted graph where the edge $\{a, b\}$ has weight 5.

$$\begin{pmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{pmatrix} \qquad \begin{pmatrix} 0 & 5 & 0 & 0 \\ 5 & 0 & 6 & 0 \\ 0 & 6 & 0 & -3 \\ 0 & 0 & -3 & 0 \end{pmatrix}$$

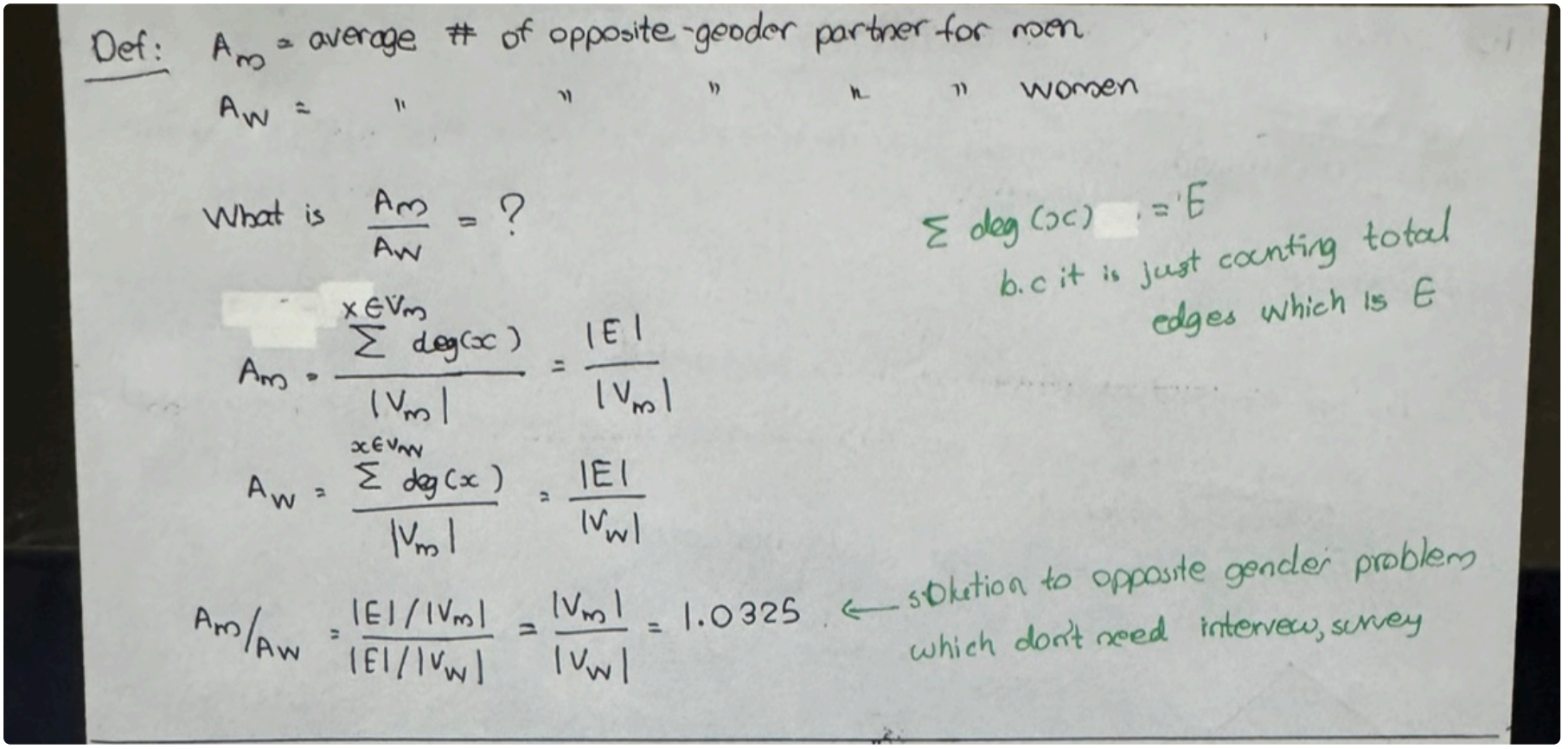
(a) (b)

Figure 5.9 Examples of adjacency matrices. (a) shows the adjacency matrix for the graph in Figure 5.6(a) and (b) shows the adjacency matrix for the weighted graph in Figure 5.8. In each case, we set $v_1 = a$, $v_2 = b$, $v_3 = c$, and $v_4 = d$ to construct the matrix.

5.2 Matching Problems

the techniques we will develop apply to much more general scenarios where instead of matching men to women, we need to match packets to paths in a network, applicants to jobs, or Internet traffic to web servers.

5.2.1 Sex in America



Collectively, males and females have the same number of opposite gender partners, since it takes one of each set for every partnership, but there are fewer males, so they have a higher ratio

The Handshaking Lemma

Lemma 5.2.1 (The Handshaking Lemma)**.** *The sum of the degrees of the vertices in a graph equals twice the number of edges.*

Proof. Every edge contributes two to the sum of the degrees, one for each of its endpoints. ■

5.2.2 Bipartite Matchings

Definition 5.2.2. A bipartite graph is a graph together with a partition of its vertices into two sets, L and R, such that every edge is incident to a vertex in L and to a vertex in R.

matching problem

In this problem, a matching will mean a way of assigning every man to a woman so that different men are assigned to different women, and a man is always assigned to a woman that he likes.

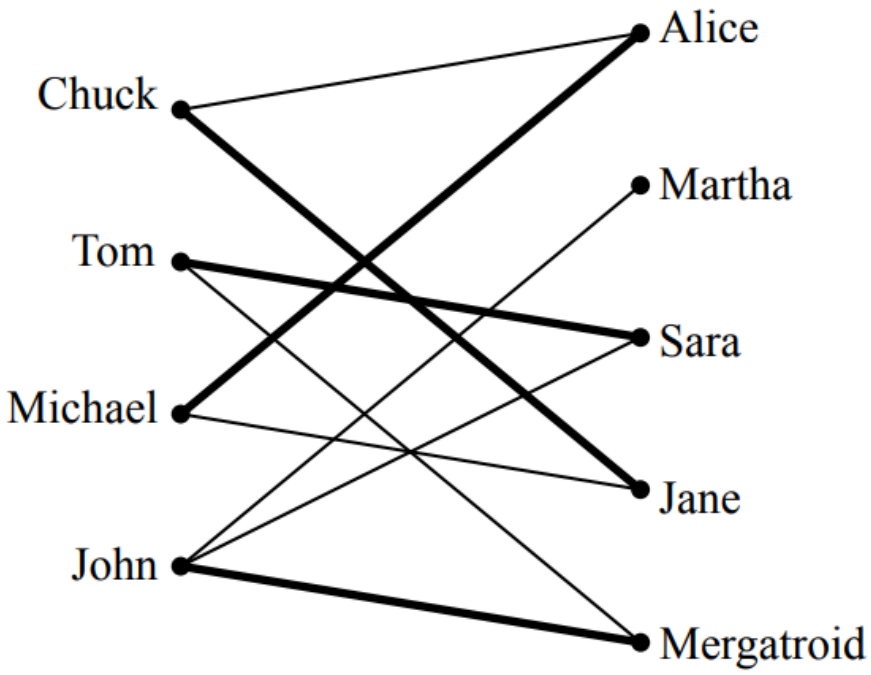


Figure 5.12 One possible matching for the men is shown with bold edges. For example, John is matched with Jane.

The Matching Condition

we will use Hall's Matching Theorem gives necessary and sufficient conditions for the existence of a matching in a bipartite graph.

the following matching condition must hold: **Every subset of men likes at least as large a set of women.**

Theorem 5.2.3

Theorem 5.2.3. *A matching for a set of men M with a set of women W can be found if and only if the matching condition holds.*

Proof. First, let's suppose that a matching exists and show that the matching condition holds. Consider an arbitrary subset of men. Each man likes at least the woman he is matched with. Therefore, every subset of men likes at least as large a set of women. Thus, the matching condition holds.

Next, let's suppose that the matching condition holds and show that a matching exists. We use strong induction on $|M|$, the number of men, on the predicate:

$$P(m) ::= \text{for any set of } m \text{ men } M, \text{ if the matching condition holds for } M, \text{ then there is a matching for } M.$$

Base Case ($|M| = 1$): If $|M| = 1$, then the matching condition implies that the lone man likes at least one woman, and so a matching exists.

Inductive Step: We need to show that $P(m)$ IMPLIES $P(m+1)$. Suppose that $|M| = m + 1 \geq 2$.

Case 1: Every proper subset[8] of men likes a *strictly larger* set of women. In this case, we have some latitude: we pair an arbitrary man with a woman he likes and send them both away. The matching condition still holds for the remaining men and women since we have removed only one woman, so we can match the rest of the men by induction.

Case 2: Some proper subset of men $X \subset M$ likes an *equal-size* set of women $Y \subset W$. We match the men in X with the women in Y by induction and send them all away. We can also match the rest of the men by induction if we show that the matching condition holds for the remaining men and women. To check the matching condition for the remaining people, consider an arbitrary subset of the remaining men $X' \subseteq (M - X)$, and let Y' be the set of remaining women that they like. We must show that $|X'| \leq |Y'|$. Originally, the combined set of men $X \cup X'$ liked the set of women $Y \cup Y'$. So, by the matching condition, we know:

$$|X \cup X'| \leq |Y \cup Y'|$$

We sent away $|X|$ men from the set on the left (leaving X') and sent away an equal number of women from the set on the right (leaving Y'). Therefore, it must be that $|X'| \leq |Y'|$ as claimed.

5.3 Coloring

5.3.1 An Exam Scheduling Problem

The Schedules Office's problem is easy to describe as a graph. There will be a vertex for each course with a final exam, and two vertices will be adjacent exactly when some student is taking both courses.

what is looked like :

- Assigning an exam to a time slot is then equivalent to coloring the corresponding vertex.
- The main constraint is that adjacent vertices must get different colors
- we should try to color all the vertices using as few different colors as possible.

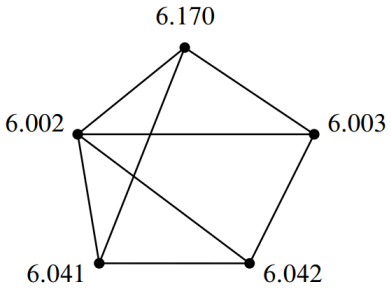


Figure 5.15 A scheduling graph for five exams. Exams connected by an edge cannot be given at the same time.

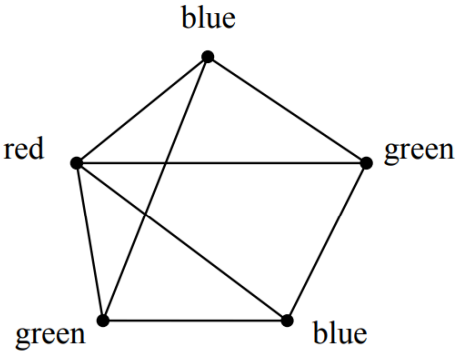


Figure 5.16 A 3-coloring of the exam graph from Figure 5.15.

you need at least 3 colors due to triangle (`6.170` , `6.002` , `6.041`)

This is an example of a graph coloring problem: given a graph G, assign colors to each node such that adjacent nodes have different colors.

- A color assignment with this property is called a *valid coloring* of the graph—a "*coloring*," for short.
- A graph G is *k-colorable* if it has a coloring that uses at most k colors.

Definition 5.3.1. The minimum value of k for which a graph G has a valid k-coloring is called its *chromatic number*, $\chi(G)$.

5.3.2 Degree-Bounded Coloring

There are some simple graph properties that give useful upper bounds on the chromatic number

bipartite -> 2 colors
planar -> 4 colors

Theorem 5.3.2.

Theorem 5.3.2. A graph with maximum degree at most k is $k-1$ colorable.

Induction parameter for graph

In graphs, typical good choices for the induction parameter are n, the number of nodes, or e, the number of edges.

Theorem 5.3.2. A graph with max degree at most k is (k+1)-colorable

Proof: by induction

P(n) ::= an n-vertex graph with max degree at most k is (k+1)-colorable

Base case (n=1):

P(1) => 0 edges

↙ ↘

d=0 1 color = d+1

Inductive step:

Assume P(n) is true for induction

Let $G = (V, E)$ be any (n+1) notes graph

Let d = max degree in G

Remove a note from G to create $G' = (V', E')$

G' has the max degree $\leq d$ & "n" nodes by P(n) G' is (k+1) colorable

Put the removed note which can have degree of $\leq d$

$\exists$ color in $\{c_1, c_2, \ldots, c_{d+1}\}$ which are not used by any neigbors

Give that color to removed one.

$\therefore$ G is (k+1) colorable.

but it have weakness since it only show upper-bond

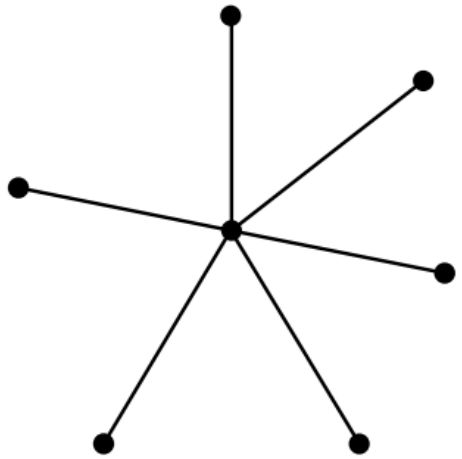


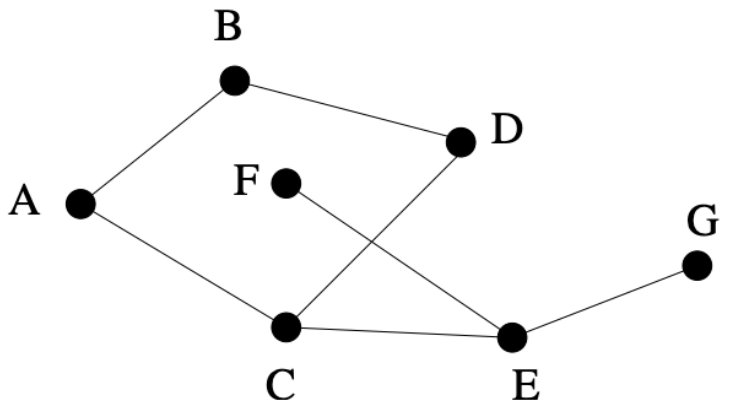
Figure 5.17 A 7-node star graph.

But sometimes $k + 1$ colors is far from the best that you can do. For example, the n-node *star graph* shown in Figure 5.17 has maximum degree $n - 1$ but can be colored using just 2 colors.

Recitation : Basic of a graph

1 Graph Basics

Let $G = (V, E)$ be a graph. Here is a picture of a graph.



Recall that the elements of V are called vertices, and those of E are called edges. In this example the vertices are $\{A, B, C, D, E, F, G\}$ and the edges are

$$\{A—B, B—D, C—D, A—C, E—F, C—E, E—G\}.$$

Deleting some vertices or edges from a graph leaves a *subgraph*. Formally, a subgraph of $G = (V, E)$ is a graph $G' = (V', E')$ where V' is a nonempty subset of V and E' is a subset of E. Since a subgraph is itself a graph, the endpoints of every edge in E' must be vertices in V'. For example, $V' = \{A, B, C, D\}$ and $E' = \{A—B, B—D, C—D, A—C\}$ forms a subgraph of G.

In the special case where we only remove edges incident to removed nodes, we say that G' is the *subgraph induced on* V' if $E' = \{(x—y | x, y \in V' \text{ and } x—y \in E\}$. In other words, we keep all edges unless they are incident to a node not in V'. For instance, for a new set of vertices $V' = \{A, B, C, D\}$, the induced subgraph G' has the set of edges $E' = \{A—B, B—D, C—D, A—C\}$.

recitation answers

Problem 1:

Pf: by induction

$P(n)$ ::= every n-vertex graph with width, w, is $(w+1)$-coloroble.

Base case:

$P(1)$: $w = 0$ colorable = 1

Inductive step:

Assume $P(n)$ is true for induction

Let $G = (V, E)$ be any $(n+1)$ notes, with width, w

1. remove the last vertex, v_{n+1} to get G' with n vertex with width w
2. by $P(n)$, G' is $w+1$ coloroble
3. add back v_{n+1} which have w, width
4. since we have $[w+1]$ color but a color from $(w+1)$ that is not color of neigborhood of v_{n+1}

$P(n+1)$ ✓ □ $\forall n \geq 1$, $P(n)$ is True

Problem 2:

(a) Theorem : Any subgraph of a planar graph is planar

Proof :

Let $G = (V, E)$ be any planar graph

Let $G' = (V', E')$ be any subgraph of G, meaning $V_0 \subseteq V$ & $E_0 \subseteq E$

since subgraph G' doesn't add new vertices and edges,

the resulting of G' has zero edge crossings.

$\therefore G'$ is a planar $\square$

(2)

Theorem : Any planner graph can be colored with at most color.

Pf: by induction

P(n) ::= Any n-vertex plannar graph can be 6-colored

Base Case:

P(1) = 1 ✓

Inductive step:

for $\forall n \geq 1$, $P(n) \rightarrow P(n+1)$

Assume P(n) = True & G_n = any planner graph with n+1 nodes

take node v with at most degree 5 and remove it from G_n to get G' subgraph.

G' is colorable by 6 colors.

Add the v node back with at most degree 5. choose one color that left

P(n+1) ✓

By the principle of induction, $\forall n \geq 1$ P(n) is true.